Neelesh Gupta

Machine Learning Researcher

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Summary

Machine Learning Researcher specializing at the intersection of Computer Vision, Geometric Deep Learning, and AI for Science applied to complex physical and biological systems, currently developing continual learning agents.

Publications

- Rohit Salla, **Neelesh Gupta**, et al.; TopoFuse: Topology-Aware Tri-Planar Fusion for 3D Cryo-Electron Tomography Segmentation (Accepted at the European Conference on Computer Vision (ECCV) 2026).
- Brahmjit Singh, **Neelesh Gupta**, et al.; Jamming Detection of Global Navigational Satellite System utilizing Deep Learning (Accepted at the Global Conference on Wireless Computing and Networking (GCWCN) 2025).

Education

National Institute of Technology, Kurukshetra, India — Nov 2022 - May 2026
B.Tech in Electrical Engineering — *CGPA: 7.04/10.00*

12th Standard - Indian School Certificate (ISC) — May 2019 - March 2021
Don Bosco Park Circus, Kolkata, India — *Percentage: 90.75%*

10th Standard - Indian Certificate of Secondary Education (ICSE) — May 2017 - March 2019
Don Bosco Park Circus, Kolkata, India — *Percentage: 91.6%*

Experience

SapientPriors | Bangalore, India — August 2026 - Present
Machine Learning Researcher

- Conducting foundational research and developing models in continual learning paradigms to build highly adaptable, autonomous AI agents for enterprise environments.

Indian Institute of Technology Roorkee | Roorkee, India — January 2026 - June 2026
Undergraduate Researcher | **Advisor:** Dr. Parikshit Pareek — *Certificate*

- Designed a hybrid neural architecture with differentiable power-balance and reserve-repair layers enforcing 100% feasibility of grid constraints; encoded spatial structure via Graph Laplacian eigenvectors and extracted Locational Marginal Prices through a KKT-based module.

Carnegie Mellon University | Pittsburgh, USA — May 2025 - Feb 2026
Undergraduate Researcher | **Advisors:** Dr. Min Xu, Dr. Xingjian Li

- Designed TopoFuse, a 3D volumetric segmentation framework that maps 2D features from a fine-tuned SAM ViT-B encoder into a 3D grid via topology-conditioned Feature-wise Linear Modulation (FiLM) fusion to weigh orthogonal view reliability against missing-wedge artifacts.

OpenNLP Labs | California, USA — April 2025 - July 2025
Machine Learning Intern — *Certificate*

- Architected an inference pipeline for a fine-tuned QwenVL model utilizing vLLM, continuous batching, and PagedAttention. Implemented weight quantization to minimize memory overhead, achieving a 38ms inference latency and outperforming RetinaNet object detection baselines.

Woosong University | Daejeon, South Korea — May - July 2025 | May - July 2024
Undergraduate Researcher (Two stints) | **Advisor:** Dr. Surender Reddy Salkuti — *Certificate*

- Built a spatio-temporal GNN with Topological Data Analysis features for solar-irradiance forecasting, achieving a 72.8% reduction in voltage violations in PV systems; fused graph convolutions with an ARIMA-based temporal model for predictive grid-stability control.

Indian Space Research Organisation (ISRO) | India — June 2024 - May 2025
Undergraduate Researcher | **Advisor:** Dr. Brahmjit Singh — *Certificate*

- Simulated a 10,000-sample GNSS dataset to systematically validate Bi-LSTM architectures for identifying and classifying radio frequency jamming. Processed time-series data utilizing overlapping sliding windows for feature extraction, achieving >99% classification accuracy across dynamic 2dB to 20dB SNR environments.

Projects

Generative Diffusion Model | PyTorch, Diffusers — *Github Link*

- Implemented a Denoising Diffusion Probabilistic Model (UNet-based forward/reverse process) for image synthesis, achieving a Fréchet Inception Distance (FID) of 12.37.

Technical Skills

Languages: Python, Java
Frameworks/Libraries: PyTorch, PyTorch Geometric, TensorFlow, Hugging Face, vLLM, OpenCV
Research Areas: Vision Language Models, Generative Models, Graph Neural Networks, Computer Vision

Achievements

- Awarded competitive research grant by the Regional Academic Centre for Space (RAC-S), ISRO (sanctioned by the Department of Space, Govt. of India) to fund independent research project.
- Top 2% (98th percentile) in Graduate Aptitude Test in Engineering(GATE) Data Science & AI 2026 paper.
- Founding member, Anveshan Anusandhan Research Cell (AARC), NIT Kurukshetra.